ADISH VASUDEVA

AI / ML ENGINEER • AGENTIC AI • RAG SYSTEMS

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Profile Summary

- Results-oriented AI/ML Engineer with 8+ years of experience designing, building, and deploying end-to-end LLM-powered agentic systems and RAG pipelines across enterprise-scale automotive and technology environments
- Strong Python expertise in building autonomous agents, RAG services, data pipelines, evaluation harnesses, and production-grade microservices using FastAPI, LangChain, LlamaIndex, and LangGraph
- Proven track record in constructing production-grade RAG pipelines with vector databases (pgvector, Pinecone, Weaviate), achieving 92% factual accuracy with sub-200ms p95 latency through hybrid retrieval and reranking
- Experienced in agent orchestration including planning loops, tool calling, memory modules, multi-step reasoning, and autonomous workflow execution using LangGraph, Semantic Kernel, and AutoGen
- Led teams of 6–8 engineers, mentoring on AI-native development practices, improving delivery timelines by 25% and reducing production defects by 30% through cross-functional collaboration
- Skilled in deploying AI services on Azure/AWS/GCP with Docker, Kubernetes, CI/CD pipelines, and monitoring for model drift, hallucination detection, and system health

Core Competencies

Agentic AI Systems	RAG Pipelines	Vector Databases	Prompt Engineering
Python (Advanced)	LLM Integration	Cloud AI (Azure/AWS)	CI/CD & MLOps
Model Fine-Tuning	Hybrid Retrieval	Eval Frameworks	Docker / Kubernetes

Work Experience

Technical Lead – AI Solutions | Jan 2024 – Present

KPIT Technologies | Bengaluru

- Lead the design and development of agentic AI systems including agent orchestration, planning loops, tool calling, and memory modules for automotive software platforms, achieving 35% improvement in system efficiency
- Architect end-to-end RAG pipelines: document ingestion, adaptive chunking, embedding generation (OpenAI, Cohere), vector indexing in Pinecone/pgvector, hybrid retrieval with BM25 + semantic search, and Cohere reranker for grounding
- Build autonomous and semi-autonomous agents for retrieval, summarization, decision support, and workflow automation using LangChain, LlamaIndex, LangGraph, and custom multi-step reasoning workflows
- Integrate enterprise-grade foundation models (Azure OpenAI GPT-4, AWS Bedrock) and transformer APIs into scalable Python/FastAPI microservices handling 10K+ daily inference requests with retry and fallback strategies
- Implement custom LLM evaluation frameworks: offline/online evals, regression test suites, content safety checks, red-team adversarial scenarios, and hallucination detection pipelines reducing false outputs by 60%
- Build monitoring infrastructure using Grafana and Prometheus for model drift detection, agent failure modes, retrieval quality metrics, latency tracking, and end-to-end system health
- Mentor a team of 6–8 engineers on AI-native development, improving delivery timelines by 25%; collaborate with product, data science, and platform teams to reduce production defects by 30%

Senior Software Engineer – AI/ML | Jan 2021 – Dec 2023

KPIT Technologies | Bengaluru

- Developed scalable AI/ML solutions using Python, achieving 40% improvement in application performance and 50% reduction in inference response time for real-time automotive diagnostic services
- Built and maintained LLM toolchains including custom prompt templates, output parsers, evaluators, and multi-step chain-of-thought reasoning workflows for enterprise automotive knowledge management
- Integrated vector databases (pgvector, Pinecone, ChromaDB) for semantic search across automotive specification documents, implementing hybrid retrieval strategies improving search relevance by 45%
- Designed and deployed containerized Python/FastAPI microservices with Docker, integrating event streams (Kafka), API gateways, and cloud-native patterns on Azure (AKS, Functions)
- Implemented rigorous code review processes and coding standards (PEP8, type hints, docstrings), reducing production defects by 35% and improving deployment success rate from 82% to 96%
- Built CI/CD pipelines (GitHub Actions, Jenkins) for AI services with safe rollouts, model versioning, feature flags, and automated regression testing enabling zero-downtime deployments
- Optimized retrieval quality and latency using freshness pipelines, reranking models, retrieval evaluators, and A/B testing frameworks for production RAG systems

Software Engineer – Automotive Test Automation | Jan 2019 – Dec 2020

General Motors | Bengaluru

- Developed Python-based automation scripts and test frameworks for Hardware-in-the-Loop (HIL) systems, automating manual testing processes for automotive ECU validation and reducing test execution time by 40%
- Built automated test report generation and documentation pipelines, processing large-scale test data to generate actionable insights and compliance reports for diagnostic testing modules
- Implemented CI/CD workflows (Jenkins, Git) for automated build, test, and deployment of test scripts; created Docker containers for reproducible test environments across HIL setups
- Developed code analysis and coverage check tools to ensure quality standards for diagnostic testing modules, achieving 90%+ code coverage across critical test suites
- Designed data processing pipelines in Python (pandas, NumPy) to extract, transform, and visualize test results, generating useful insights for hardware verification and validation teams
- Automated diagnostic testing workflows for automotive ECUs using UDS protocols, CAN bus communication, and custom Python libraries, reducing manual effort by 60%
- Contributed to 15+ internal knowledge-sharing sessions on test automation best practices, HIL setup configurations, and automotive use cases for hardware verification
- Collaborated with cross-functional teams (hardware, firmware, validation) to gather requirements, define test specifications, and streamline information gathering for vehicle diagnostic systems

Technical Skills

Languages: Python (Advanced), Java, JavaScript/Node.js, SQL, Bash, C/C++

AI/ML & LLM: LangChain, LlamaIndex, LangGraph, Semantic Kernel, AutoGen, PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, OpenAI API

RAG & Retrieval: Chunking strategies, OpenAI/Cohere Embeddings, Cohere Rerank, Hybrid Search (BM25 + Semantic), Document Grounding, Freshness Pipelines

Vector Databases: pgvector, Pinecone, Weaviate, Milvus, ChromaDB, FAISS

Cloud & DevOps: Azure (OpenAI, AKS, Functions), AWS (Bedrock, SageMaker, Lambda), GCP (Vertex AI), Docker, Kubernetes, GitHub Actions, Jenkins

Monitoring & Eval: RAGAS, DeepEval, pytest, Custom eval harnesses, Hallucination detection, Model drift monitoring, Grafana, Prometheus

Automotive & Tools: HIL Systems, CAN/UDS Protocols, Diagnostic Testing, FastAPI, gRPC, Kafka, Git, JIRA, Confluence

Key Projects

- **AutoDoc AI – Intelligent Document Agent:** End-to-end agentic RAG system ingesting 50K+ enterprise documents with autonomous multi-hop reasoning, citation-grounded Q&A, achieving 92% factual accuracy at <180ms p95 latency using LangChain, LangGraph, Pinecone, and Cohere Rerank
- **Automotive Knowledge Assistant:** LLM-powered conversational agent for automotive specification search and summarization, integrating hybrid retrieval (BM25 + semantic) across vector databases, serving 500+ internal engineers with real-time decision support
- **HIL Test Automation Framework:** Python-based end-to-end test automation platform for Hardware-in-the-Loop systems at General Motors, automating ECU diagnostic testing via UDS/CAN protocols, reducing manual testing effort by 60% and test execution time by 40%

Education

Master’s Degree – Artificial Intelligence & Machine Learning
Coventry University, United Kingdom
2022 – 2024

Bachelor of Engineering – Electronics and Communication
Bangalore Institute of Technology, Bengaluru
2014 – 2018

Certifications

- DeepLearning.AI – LangChain for LLM Application Development
- DeepLearning.AI – Building Agentic RAG with LlamaIndex
- DeepLearning.AI – Prompt Engineering for Developers
- Hugging Face NLP Course – Transformer Models & Fine-Tuning